KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The v6 reduction — wide calibration set, freely fitted, Gaia applied afterwards

Prepared 2026-09-09. Solution: `~/KMTdata/Results/v6/IFfinal_<field>.mat` (variable `IFsys`), with the tie in `Tie_<field>.mat`.

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1. A short description

The analysis is restricted to a calibration set defined by **magnitude and isolation alone**: 14 < I < 19 with no companion brighter than I = 18 within 2 arcsec. Every one of those stars is freely fitted and every one takes part in determining the per-epoch frame at every step. **No Gaia quality cut is applied at selection**; RUWE enters only afterwards, when the finished solution is tied to Gaia. The target passes the selection on its own merits and needs no special handling.

This is v5 with the magnitude window widened from 14–17 to 14–19 and the RUWE cut moved to the end. Both changes help: **v6 measures the target better than any previous reduction, including the reference one.**

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2. Object selection, step by step

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in 14 < I < 19	529	470
... and isolated: no I < 18 companion within 2.0 arcsec	**287**	**253**
= analysis set actually fitted	**287**	**253**

The isolation cut is now the dominant one, removing 46% of the magnitude-selected sources on both fields. The set is more than twice v5's 130 / 100.

The target is a member. At I = 18.13 it lies inside the window and passes the isolation test, so unlike v4 and v5 it is not added by hand and no "passenger" stars are needed.

For the tie only, Gaia is required: **273 of 287** and **225 of 253** sources have a Gaia counterpart, and of those **220** and **177** pass RUWE < 1.4.

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3. Gaia, applied only at the end

The map. Gaia standard coordinates are fitted directly onto cut-out pixels from the RUWE-clean stars — not carried through OGLE, which would inherit its ~110 mas astrometric error. Median residual **35.1 mas (BLG41)** and 37.1 (BLG01), with scales of 2.5198 and 2.5291 pix/arcsec against the 2.500 expected at 0.4″/pix.

The gauge, and a correction to all earlier work. A free fit determines astrometry only up to a gauge: applying a global affine to all positions and compensating in the per-epoch transformations changes no residual. In the proper motions this freedom is **a constant plus a linear gradient across the field** — six parameters, not two.

Every tie made in this project before v6 removed **only the constant**. The residual shows up plainly in the cross terms: `corr(Δμ_α, Y) = −0.75` on BLG41, a **shear of −0.037 mas/yr per pixel**, and −0.035 on BLG01 — nearly the same value in two independently reduced fields, so it is a property of the method, not noise. Removing the full six-parameter gauge:

scatter of our proper motions about Gaia	constant only	full gauge
BLG41	4.34 / 3.57 mas/yr	**0.98** / **0.66**
BLG01	3.48 / 3.20 mas/yr	**1.00** / **0.70**

A factor of three to four. The gauge removed here is a constant of +0.015 / +5.170 (BLG41) and +2.275 / +4.329 (BLG01) mas/yr, plus the gradient above. **The absolute proper motions in the earlier reports are affected by this and should be read as superseded.**

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4. The chain, and every effect fitted in it

Two steps, both on the 287 / 253 source set:

1. **Full-decade SysRem correction** — 2 components on the astrometric residuals, computed once and saved. 2. **Final solution**, reusing that correction.

	BLG41	BLG01
step 1	19.8 min	16.7 min
step 2	35.6 min	31.3 min
convergence over stars brighter than 17, last 3 iterations	0.0028 mas	0.0007 mas

Each step runs two passes internally. `IFsys.AnnualEffect` and `IFsys.PixPhase` read `0` in the saved solutions, which does **not** mean those effects were ignored: they are fitted and subtracted from the data in pass 1 and forced off in pass 2, which works on data from which they have already been removed.

	pass 1	between	pass 2 (final)
per-epoch affine	fitted		fitted
per-source position and motion	fitted		fitted
DCR / chromatic refraction	fitted		fitted again
annual term	fitted and subtracted from the data		forced off
pixel-phase term	fitted and subtracted from the data		forced off
SysRem		applied	
iterative reweighting	15 iterations		6 iterations

- **Per-epoch affine** — 6 parameters per exposure in `ParE`, stored as the deviation from the identity: translation (2), rotation (1), uniform scale (1), shear (2). Each epoch's six parameters come from 287 sources, so 574 equations for 6 unknowns, against 260 in v5 and 1188 in the reference reduction.

- **Per source** — position at `JD0` = 2019-06-01 and proper motion, 4 parameters in `ParS`. All free; `ParSFixed` is empty.
- **DCR** — `[1, sin(pa)·secz, cos(pa)·secz]` plus six higher orders in `(secz·sin pa)`^{n}, `(secz·cos pa)`^{n} for n = 2, 3, 4: 9 per axis, 18 in total, fitted in 6 equal-population colour bins. Only the *differential* part is recoverable — refraction common to the field is exactly a translation and is absorbed by `t_x, t_y` every epoch.
- **Annual term** — a quartic in the phase of the calendar year, 10 parameters, fitted globally, one curve shared by every star. Not parallax, which is per-source and switched off.
- **Pixel phase** — a quintic in the sub-pixel phase, 10 parameters, global.
- **SysRem** — 2 components over the whole decade, reused rather than refitted per season, with point-wise rejection.
- **Weighting** — each source weighted by its own residual scatter, recomputed each iteration, with moving-median outlier rejection over 30 epochs.

No per-season fitting enters the solution. Seasons are used only for reporting.

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5. Results

	BLG41 ΔX / ΔY	BLG01 ΔX / ΔY
stars brighter than I = 17	**5.15 / 5.06**	**4.93 / 5.12**
the target	**23.75 / 19.63**	**21.36 / 20.37**

mas. Against the other reductions:

target residual RMS	BLG41	BLG01
v3, reference, ~600 sources	24.62 / 21.00	25.48 / 21.78
v5, 130 / 100 sources, RUWE-cut	25.61 / 22.27	28.45 / 23.15
v6, 287 / 253 sources	**23.75 / 19.63**	**21.36 / 20.37**

v6 is the best on every axis in both fields — 3.5% better than the reference on BLG41 and 16% better on BLG01. The lesson from v5 is now clear: restricting the set to bright isolated stars helps, but restricting it *too far* costs more than it gains. 287 isolated stars beat both 130 isolated stars and ~600 unfiltered ones.

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6. Proper motion of the target, with uncertainties

	μ_α cos δ	μ_δ
BLG41	**−2.017 ± 0.335**	**−7.163 ± 0.360**
BLG01	**−1.781 ± 0.256**	**−7.136 ± 0.377**
difference between fields	−0.236 ± 0.421	−0.028 ± 0.521

mas/yr. The two fields agree **within their errors**, at 0.56 and 0.05 sigma.

The quoted error is the **internal** one: the slope uncertainty from the scatter of the ten season means about the fitted line. It is deliberately not computed from the per-epoch count, which would give an absurdly small number — the nightly noise is strongly correlated. Median internal uncertainty over all sources is 0.127 / 0.114 (BLG41) and 0.123 / 0.112 (BLG01) mas/yr.

That internal error is still optimistic by a factor of two to three. The external check — the scatter of our proper motions about Gaia's, 0.98 / 0.66 and 1.00 / 0.70 mas/yr per star — is larger than the internal estimate would predict even after Gaia's own 0.15 mas/yr is allowed for. Use ~**1.0** / **0.7 mas/yr** as the realistic per-star uncertainty; the internal figure captures the random part only and misses systematics that bias a star's slope while leaving its seasons internally consistent.

Gaia's own value for the target is −2.479 / −9.129 mas/yr. Our μ_α cos δ agrees; our μ_δ differs by about 2 mas/yr, which is 3 sigma on the external error. The target is blended with a neighbour 4.5 pix away and its motion here is fitted straight through a multi-year magnification event, so this is not a clean comparison.

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7. Comparison stars

Six, chosen to be the same physical objects in both fields — three near the target's own magnitude and three brighter:

tag	I	dist [pix]	decade RMS ΔX / ΔY, BLG41	BLG01
`m18_d04`	18.14	4.5	29.03 / 27.91	24.84 / 26.86
`m18_d17`	18.09	17.3	51.68 / 27.85	85.47 / 28.47
`m18_d48`	18.05	48.4	17.92 / 16.76	14.58 / 15.57
`m16_d24`	16.68	24.4	**4.19** / **4.27**	**4.17** / **4.38**
`m16_d30`	16.63	30.4	6.91 / 5.85	6.43 / 5.55
`m16_d31`	16.75	30.6	8.48 / 7.79	11.20 / 7.71

mas. The bright trio reproduces between the fields almost exactly — `m16_d24` gives −3.31 / −7.81 on BLG41 and −3.31 / −7.89 on BLG01 — while the I ≈ 18 trio does not: `m18_d17` gives −6.03 against −10.05 mas/yr, and its decade RMS reaches 52 and 85 mas with only ~2400 usable epochs. It is a poor star and should not be used as a control.

`m18_d04` is the neighbour blended with the target, 4.5 pix away under ~7 pix seeing. It is a legitimate member of the calibration set — the target at I = 18.13 is fainter than the I < 18 companion threshold, so it does not trigger rejection — but its centroid tracks the target's brightening through the event, and it is not an independent object.

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8. Chi-squared of the monthly binned residuals

For every star the residuals are binned in sidereal months; each bin contributes `(mean/standard error)²` in both axes, and `DoF = 2·N_bins − 4`. Values are not divided by DoF.

	BLG41	BLG01
stars measured	287	253
median DoF	180	180
target χ²	**1574** (DoF 180)	**1313** (DoF 180)

target χ^2/DoF	8.7	7.3
field median χ^2/DoF	**7.2**	**7.1**

The target is unremarkable. Its χ^2 sits in the body of the distribution, not in the tail: 8.7 against a field median of 7.2 on BLG41, and 7.3 against 7.1 on BLG01. Whatever astrometric excursion the event produces is not large enough to push the target out of the population of ordinary stars.

The field median of χ^2/DoF ≈ 7 is itself informative: it says the standard error of the mean within a month **underestimates the true bin uncertainty by about $\sqrt{7} \approx 2.6$**, because the noise within a month is correlated rather than independent. This is the same factor that makes the internal proper-motion errors of section 6 optimistic.

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9. Overlap of the two fields

Stars are identified through the OGLE catalogue, which resolves about six times more sources than KMT detects; matching the two source lists directly is ambiguous because a close pair may be split in one field and merged in the other.

	BLG41	BLG01
sources	594	621
matched to an OGLE entry	551 (93%)	497 (80%)
distinct OGLE stars behind them	**454**	**394**

OGLE I	BLG41	BLG01	in both	distinct	found in both
< 15	28	22	22	28	78.6%
15 – 16	54	42	42	54	77.8%
16 – 17	172	149	146	175	83.4%
17 – 18	135	119	117	137	85.4%
18 – 19	55	44	41	58	70.7%
all brighter than 19	**444**	**376**	**368**	**452**	**81.4%**

About four stars in five are detected in both fields, flat from the brightest down to I = 18. **There is nothing fainter than I = 19 to compare**: our source lists span 12.87 to 18.98 in both fields, while OGLE itself reaches 21.40 and holds 2660 entries fainter than 19. That is the KMT detection limit in this crowding, not a catalogue cut.

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10. Files

file	content
`report_v6_RMS.png`	residual RMS against OGLE I, per axis and field
`report_v6_pm_vs_gaia.png`	our absolute proper motion against Gaia's, after the tie
`report_v6_chi2.png`	χ^2 distribution of the monthly binned residuals, target marked

report_v6_radec_<field>_<tag>.png	RA against time, Dec against time, and RA against Dec colour-coded by time
report_v6_motion_<field>_<tag>.png	the 2x2 motion figures, proper motion retained and removed
source_motion_v6_<field>.csv	the target, per epoch
source_motion_v6_<field>_<tag>.csv	each comparison star, per epoch

CSV columns are `JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season`, with positions relative to that star's own mean and **proper motion not removed**. Headers carry the star's magnitude, pixel position, sky position and absolute proper motion.

In the motion figures the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky. **Pixel X runs opposite to RA**, so those two carry opposite signs in X.

Figures

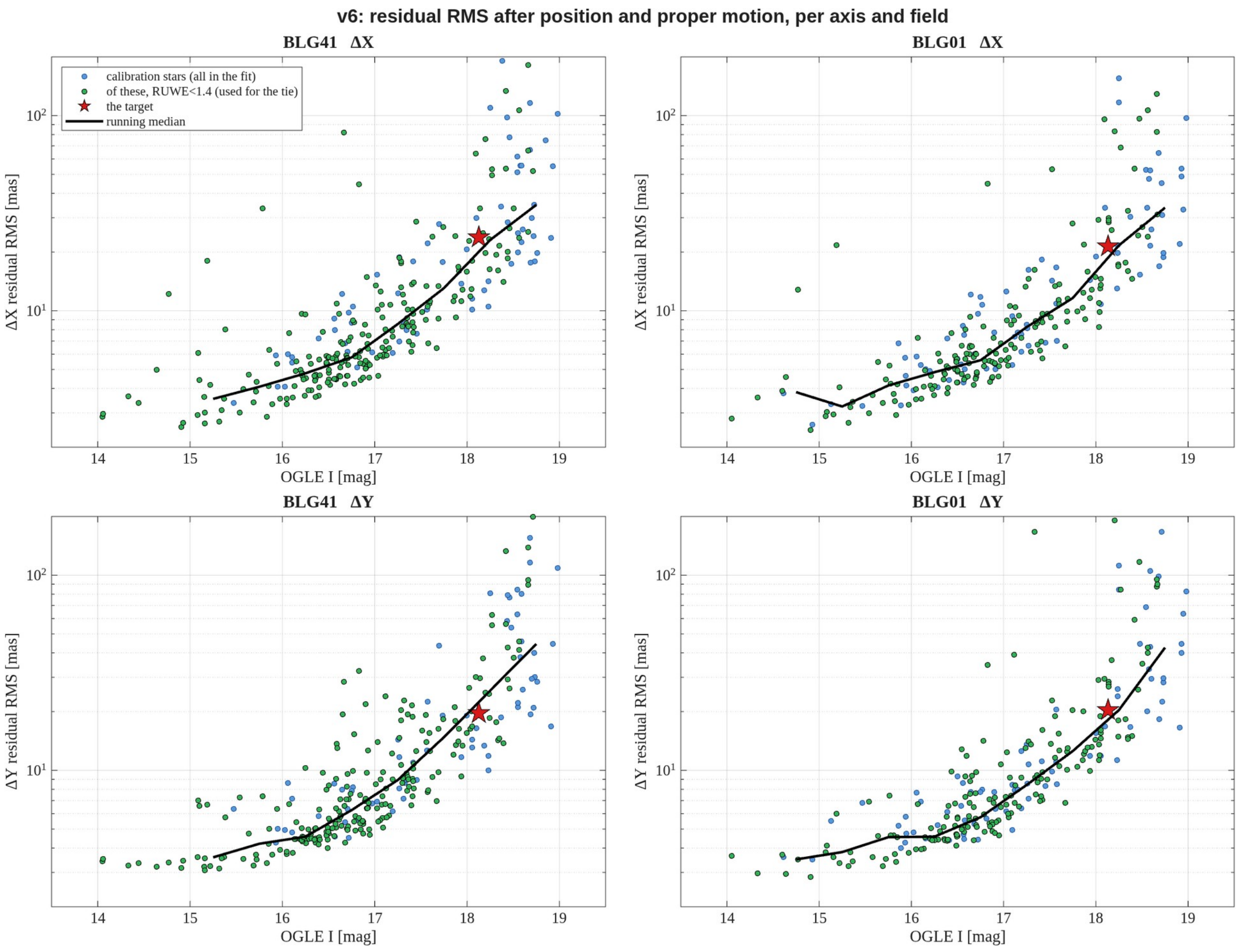

Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, after position and proper motion have been removed. Blue: all 287 / 253 calibration stars, every one freely fitted. Green: the subset with a Gaia RUWE below 1.4, used only for the tie. Red: the target. Black: running median.

`report_v6_RMS.png`

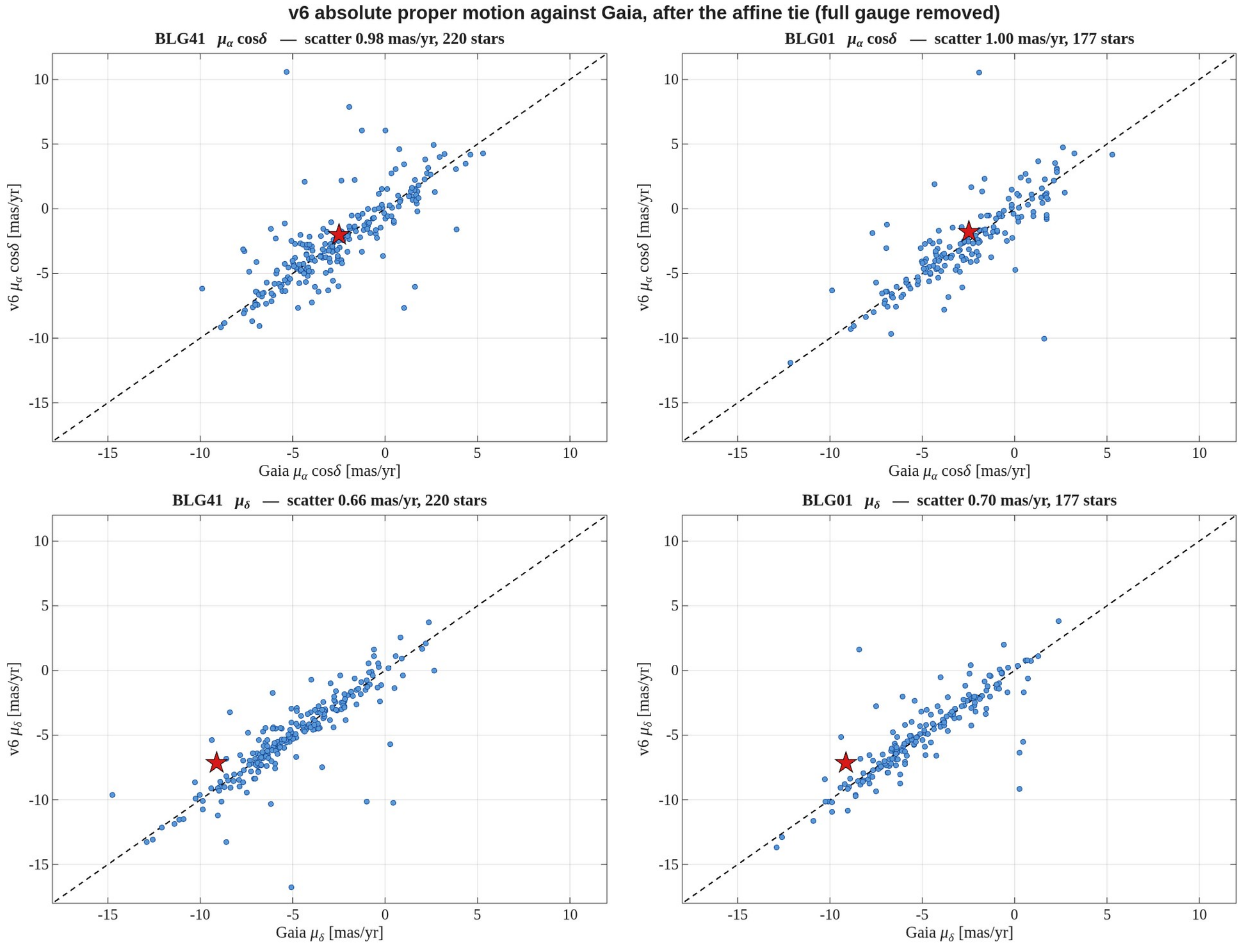

Figure 2. Our absolute proper motion against Gaia's, per axis and per field, after the affine tie with the **full six-parameter gauge** removed. Dashed line is 1:1. The scatter quoted in each panel is the robust standard deviation of the difference; it improves by a factor of three to four over a tie that removes only the constant. The target is the red star.

`report_v6_pm_vs_gaia.png`

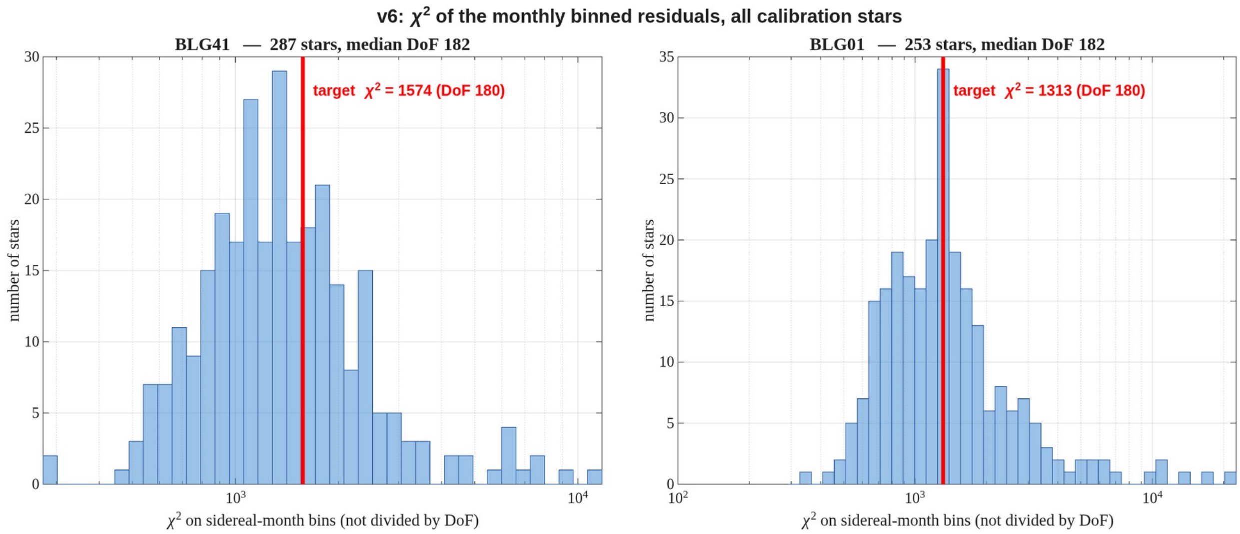

Figure 3. χ^2 of the sidereal-month binned residuals for every calibration star, both axes combined, **not divided by DoF**. Each bin contributes (mean / standard error)² per axis and DoF = $2N_{bins} - 4$. The red line marks the target, which sits in the body of the distribution rather than the tail.

`report_v6_chi2.png`

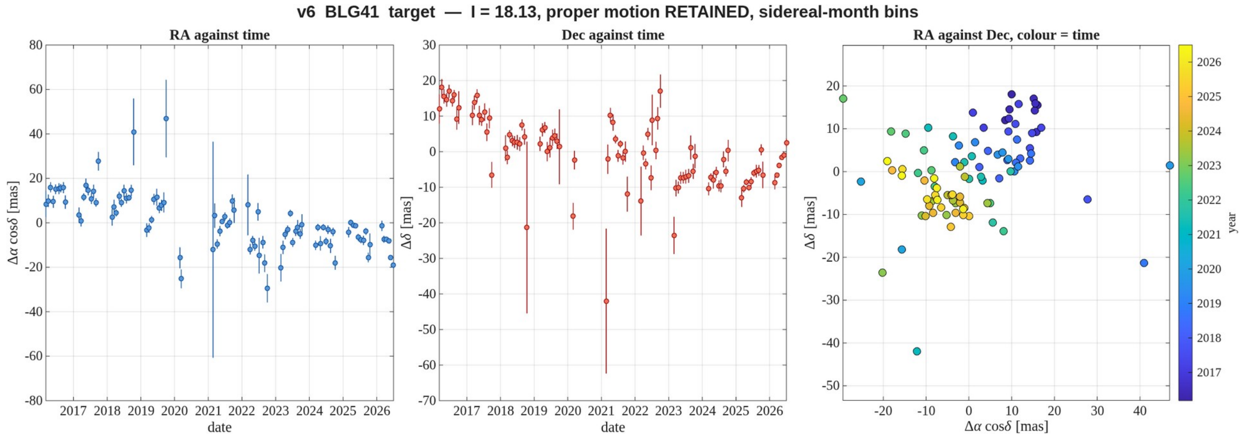

Figure 4. **BLG41** — the target, I = 18.13. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG41_target.png`

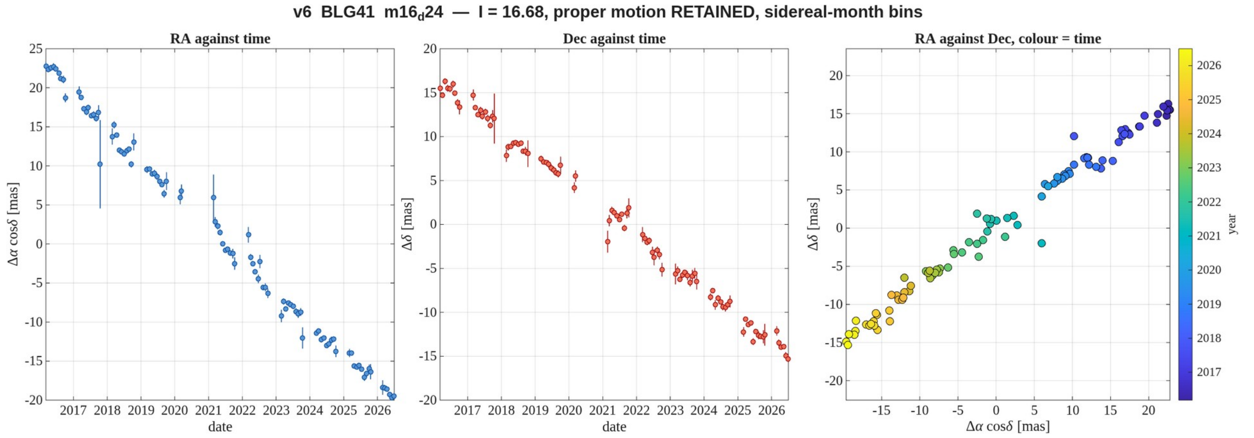

Figure 5. **BLG41** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG41_m16_d24.png`

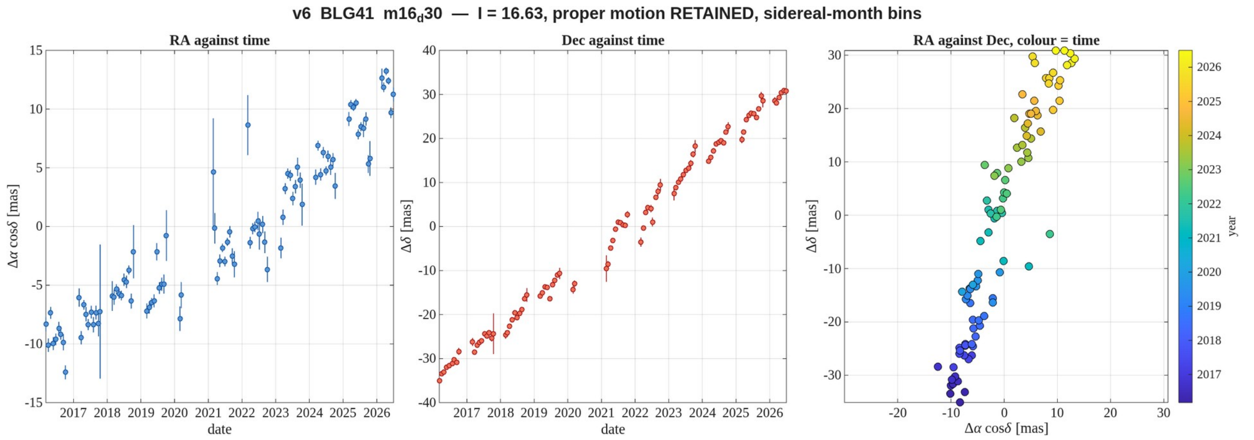

Figure 6. **BLG41** — comparison star I = 16.63, 30 pix from the target. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG41_m16_d30.png`

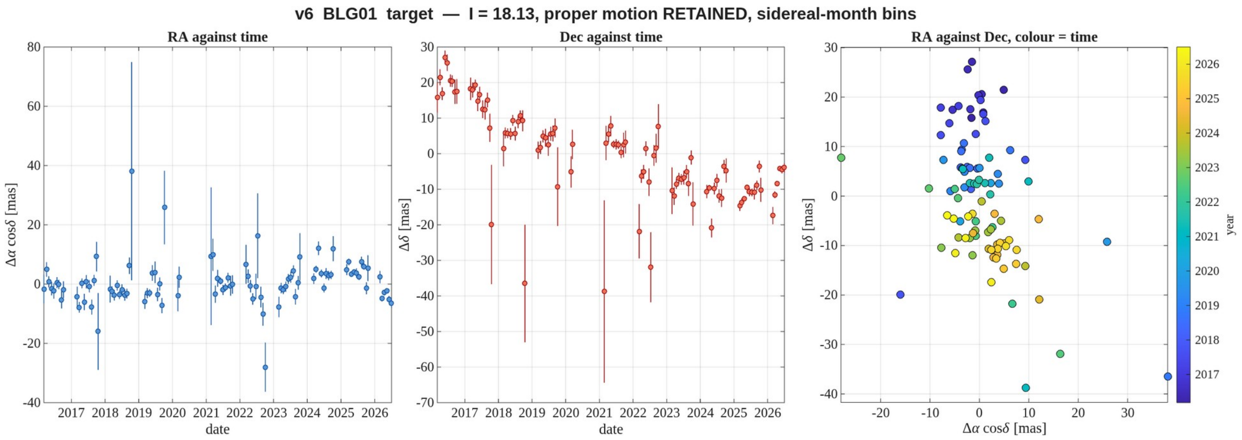

Figure 7. **BLG01** — the target, I = 18.13. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG01_target.png`

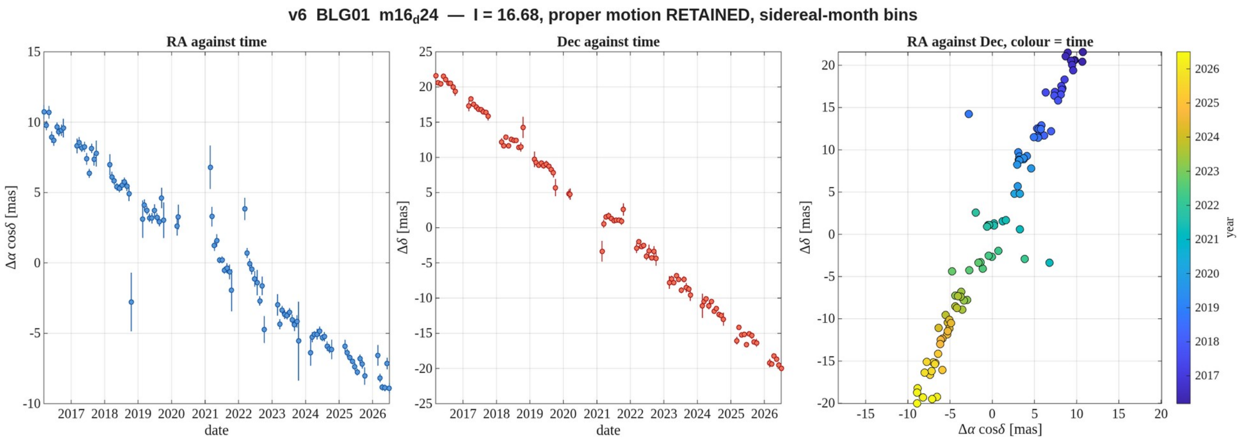

Figure 8. **BLG01** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG01_m16_d24.png`

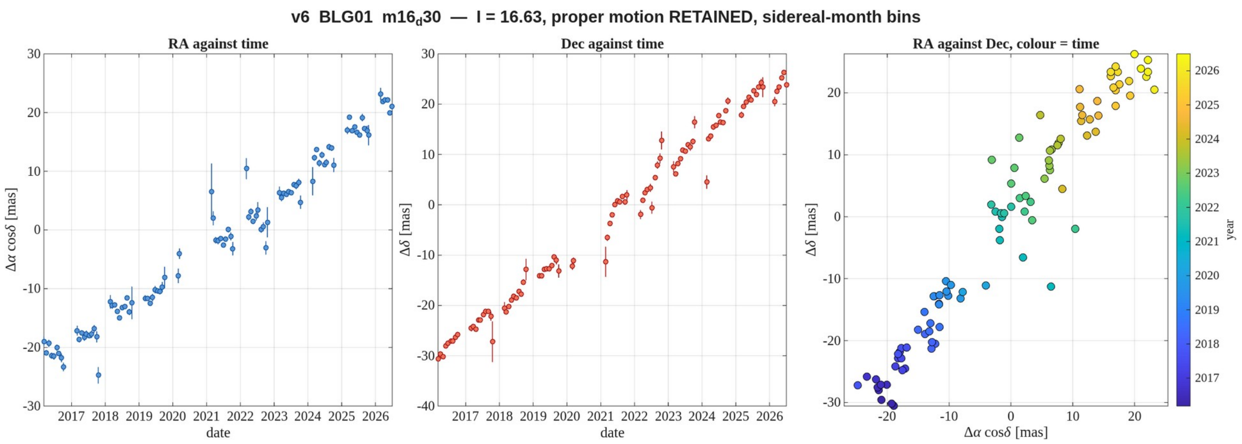

Figure 9. **BLG01** — comparison star I = 16.63, 30 pix from the target. Left: RA against time. Middle: Dec against time. Right: RA against Dec with time colour-coded, where the proper motion appears as the ordered progression of colour. Sidereal-month bins, proper motion **retained**, positions relative to the star's own mean.

`report_v6_radec_BLG01_m16_d30.png`

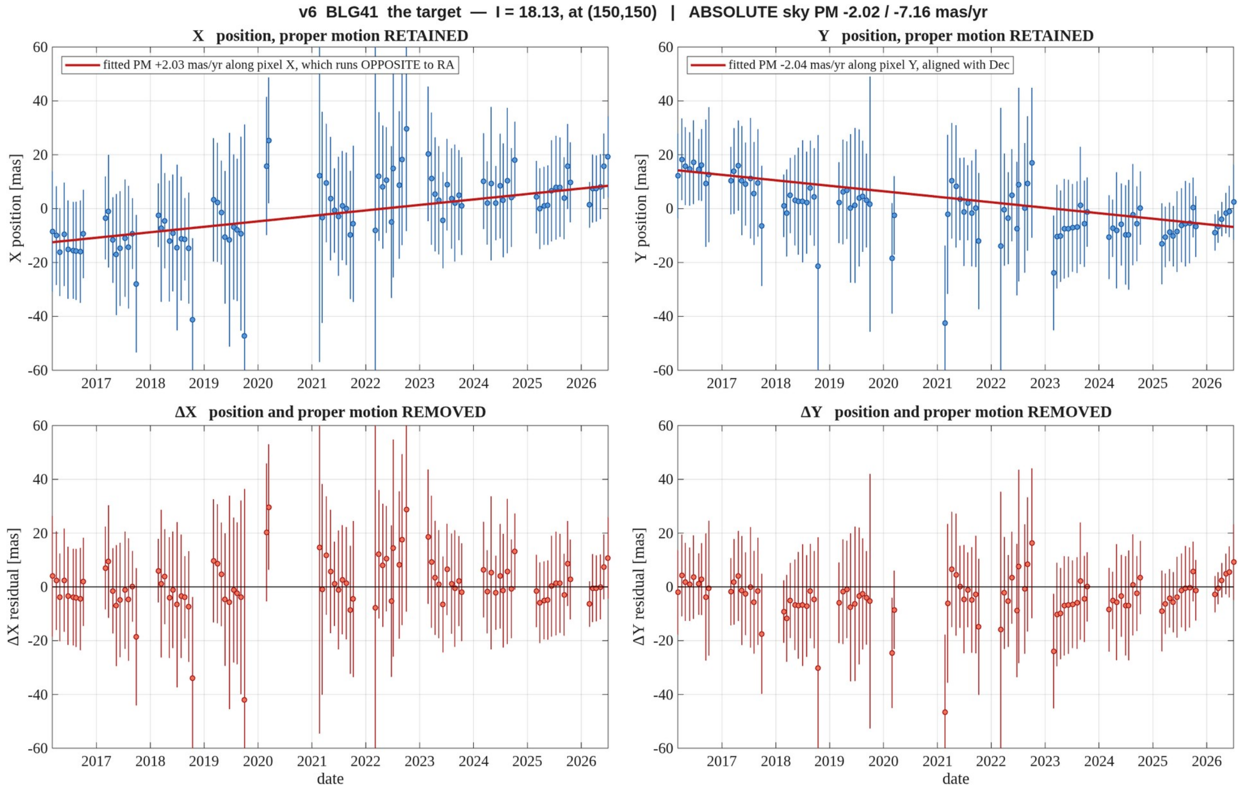

Figure 10. **BLG41** — the target, I = 18.13. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_target.png`

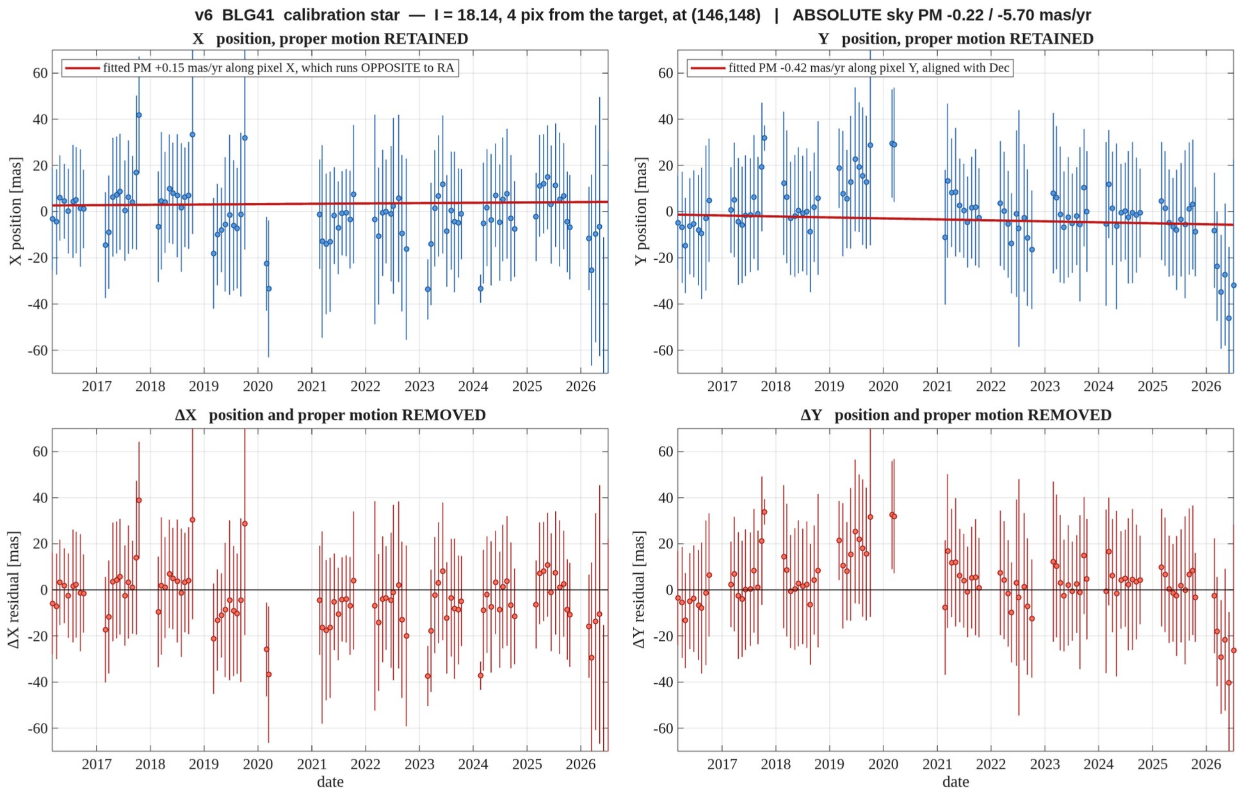

Figure 11. **BLG41** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d04.png`

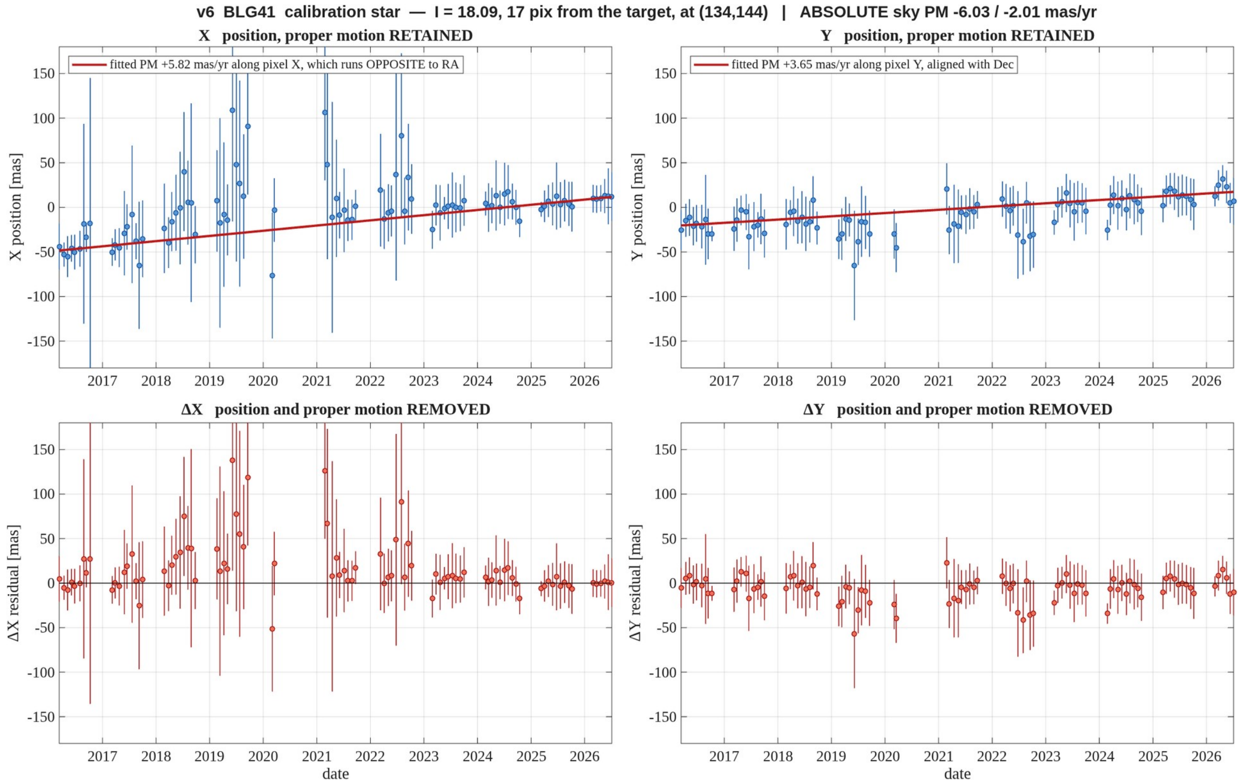

Figure 12. **BLG41** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d17.png`

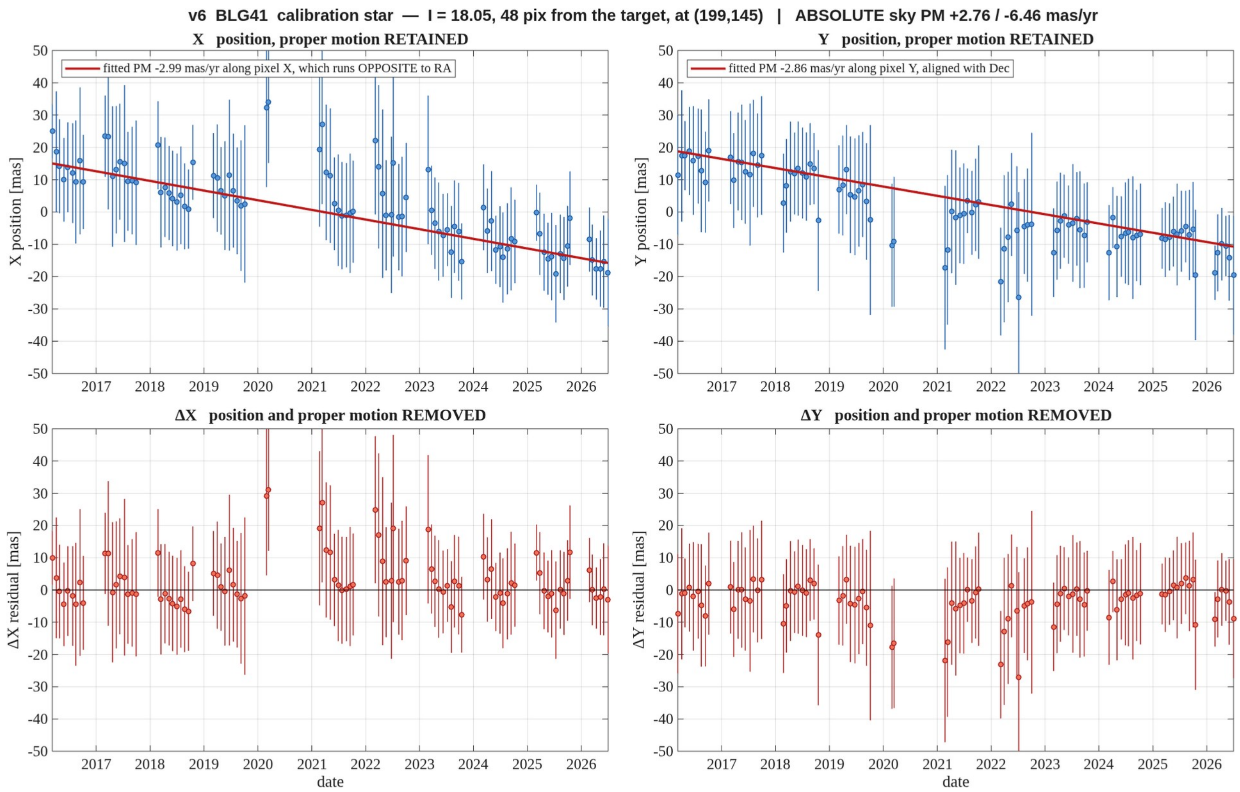

Figure 13. **BLG41** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d48.png`

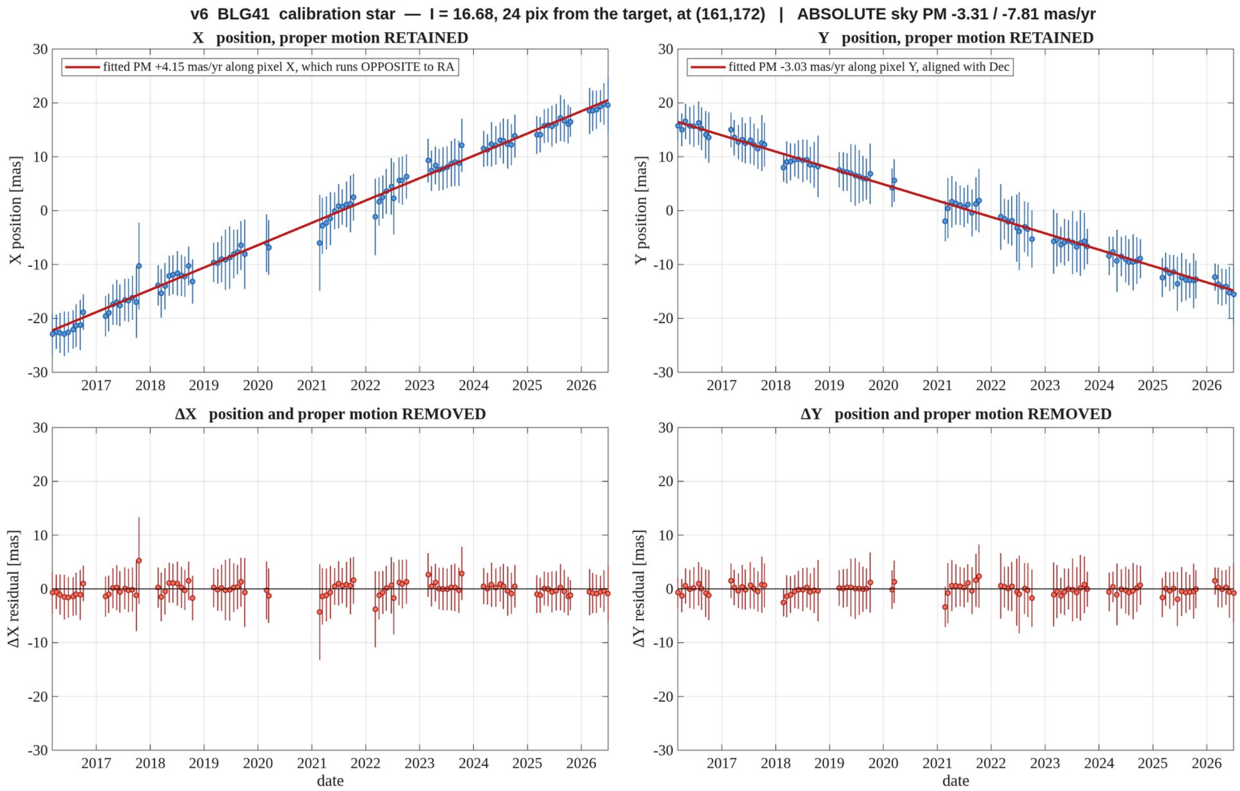

Figure 14. **BLG41** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d24.png`

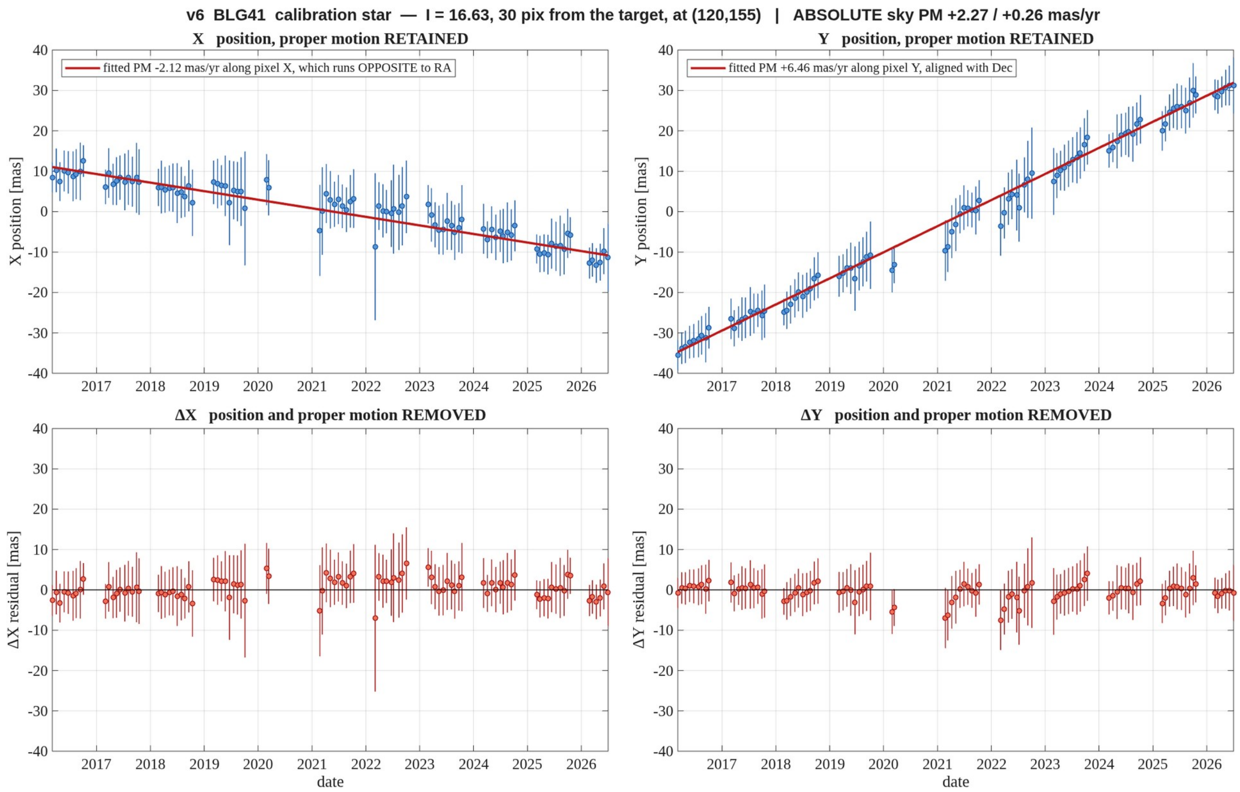

Figure 15. **BLG41** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d30.png`

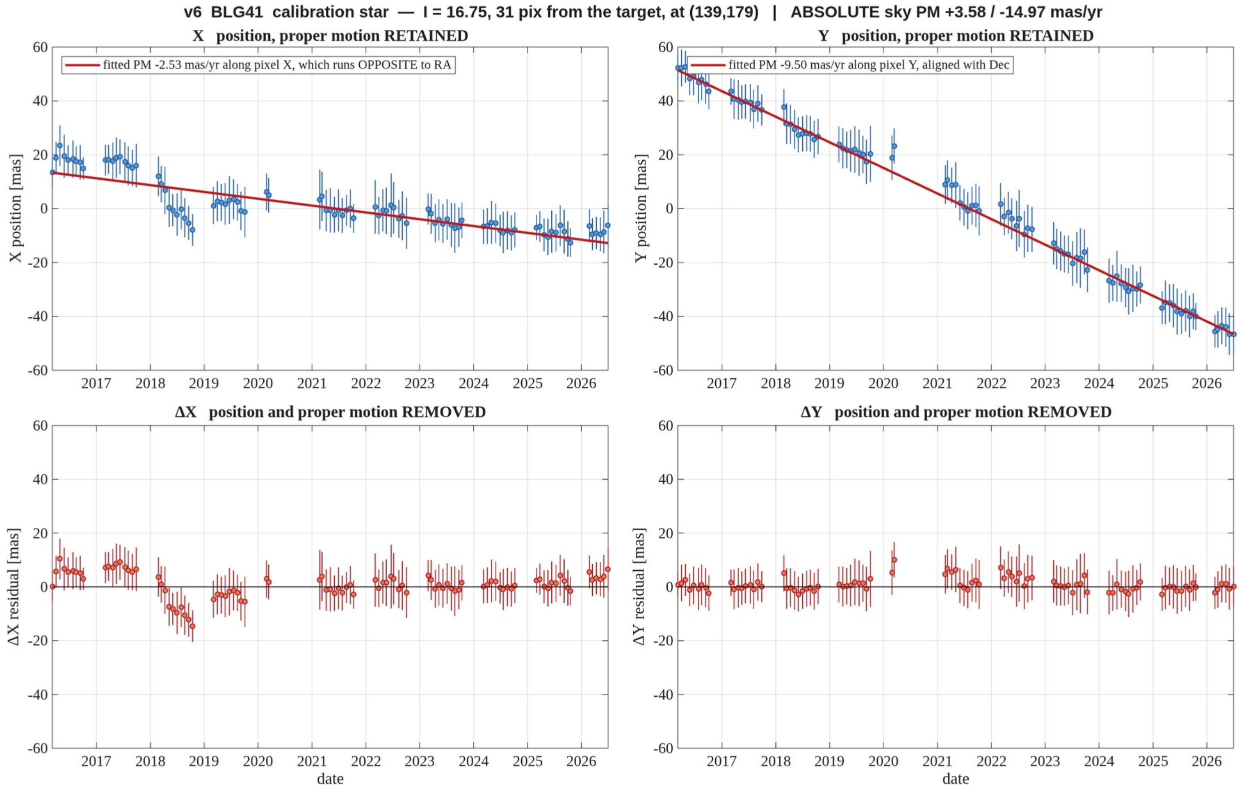

Figure 16. **BLG41** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d31.png`

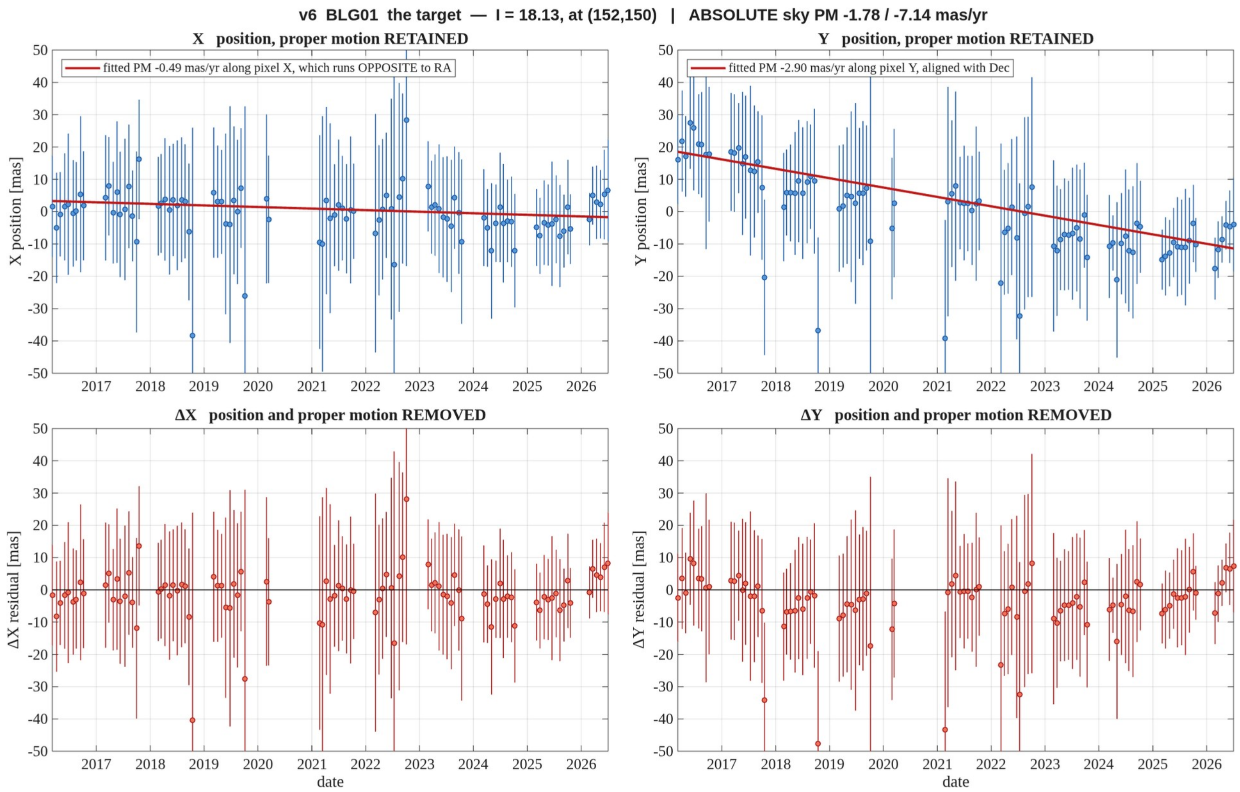

Figure 17. **BLG01** — the target, I = 18.13. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_target.png`

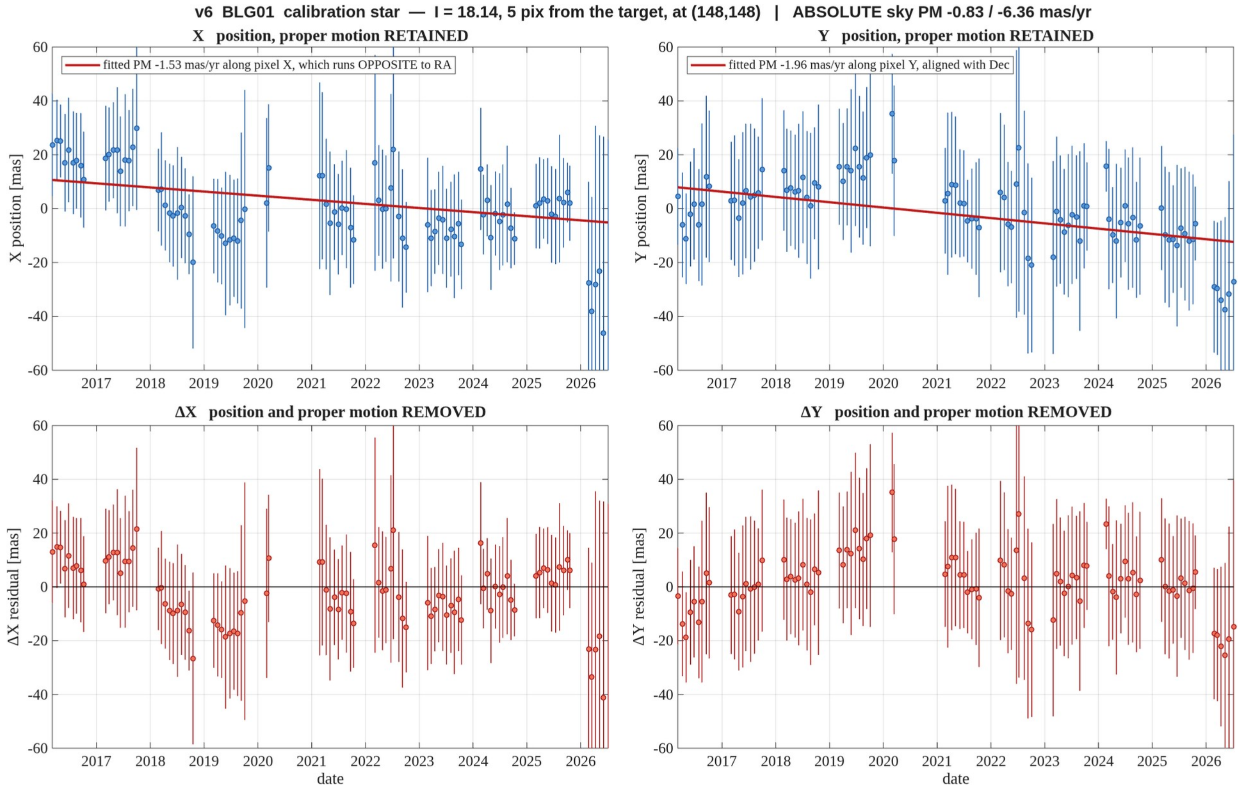

Figure 18. **BLG01** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d04.png`

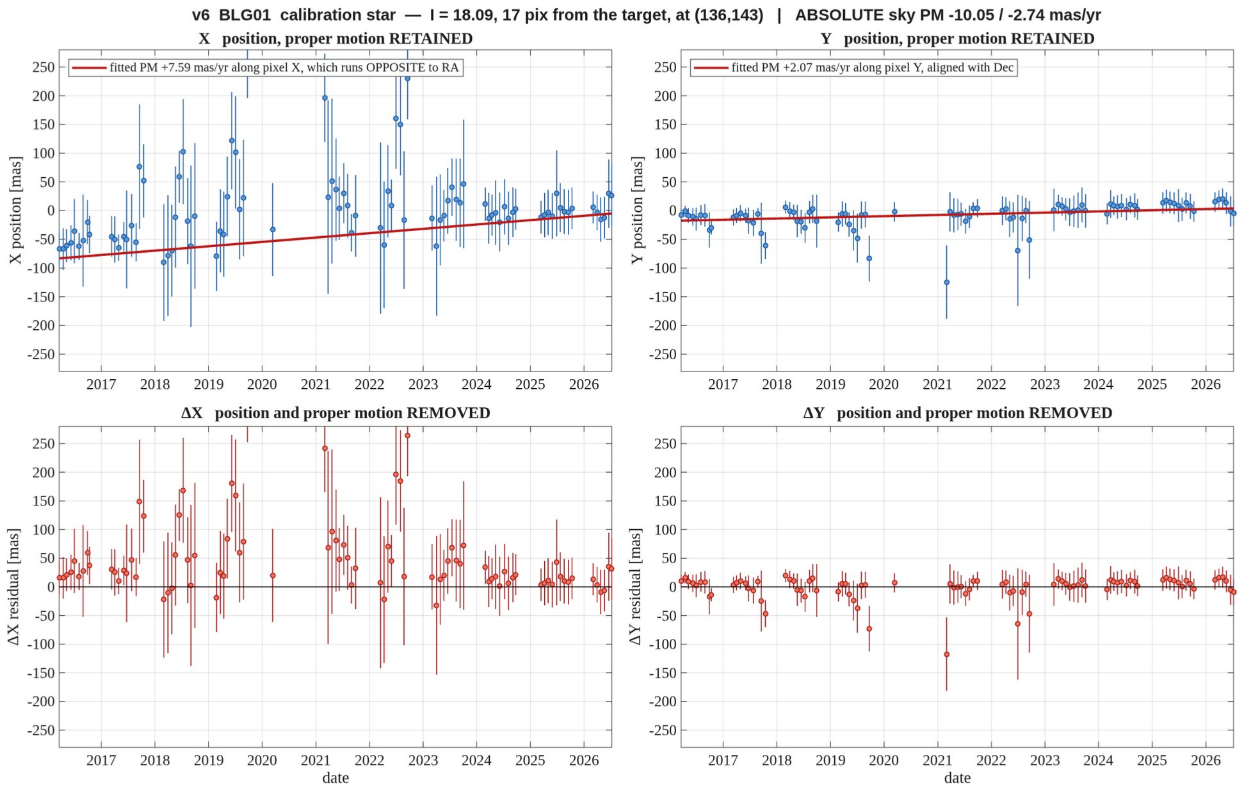

Figure 19. **BLG01** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d17.png`

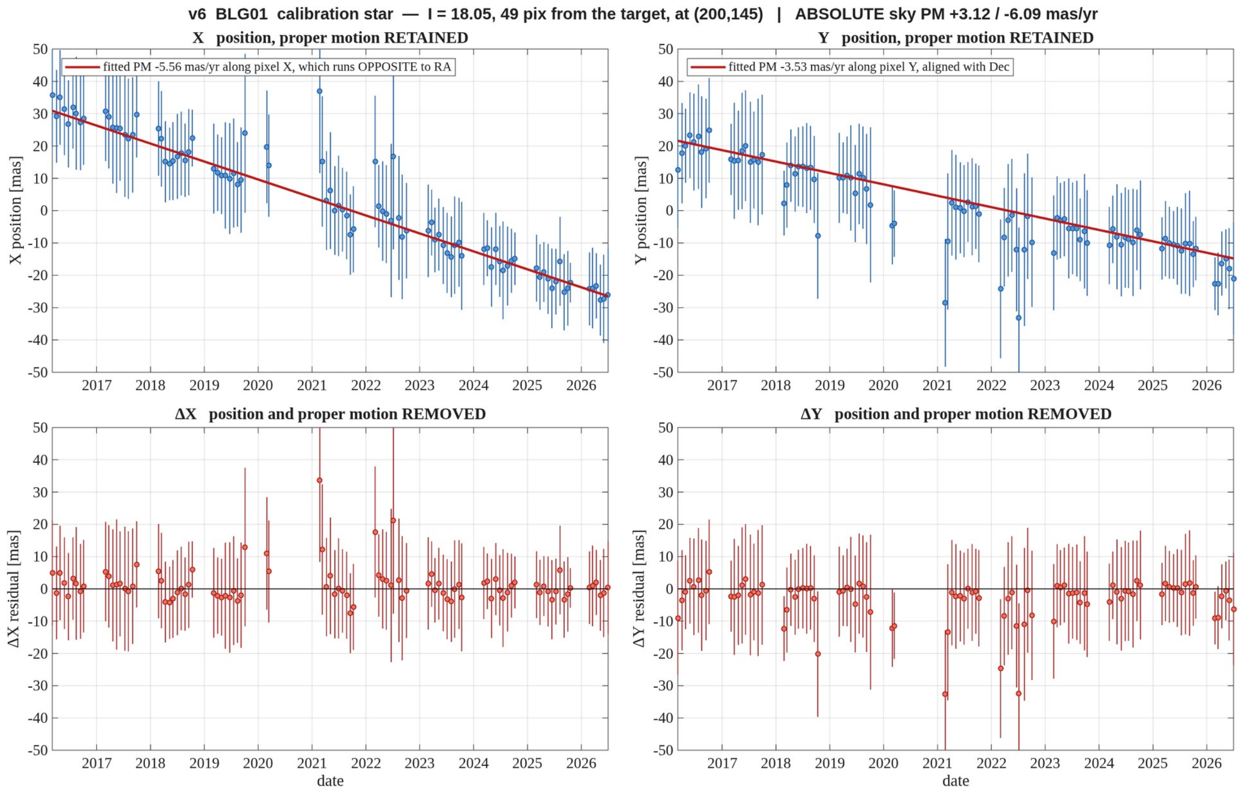

Figure 20. **BLG01** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d48.png`

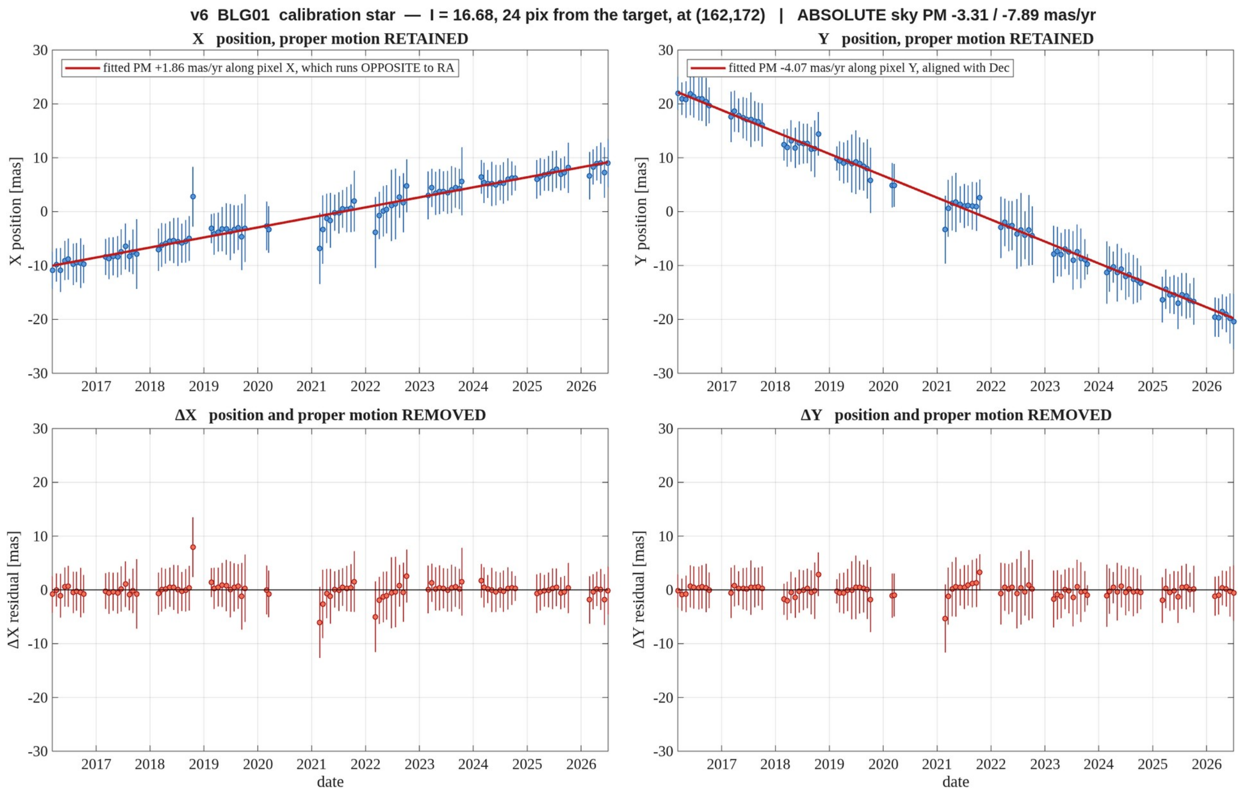

Figure 21. **BLG01** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d24.png`

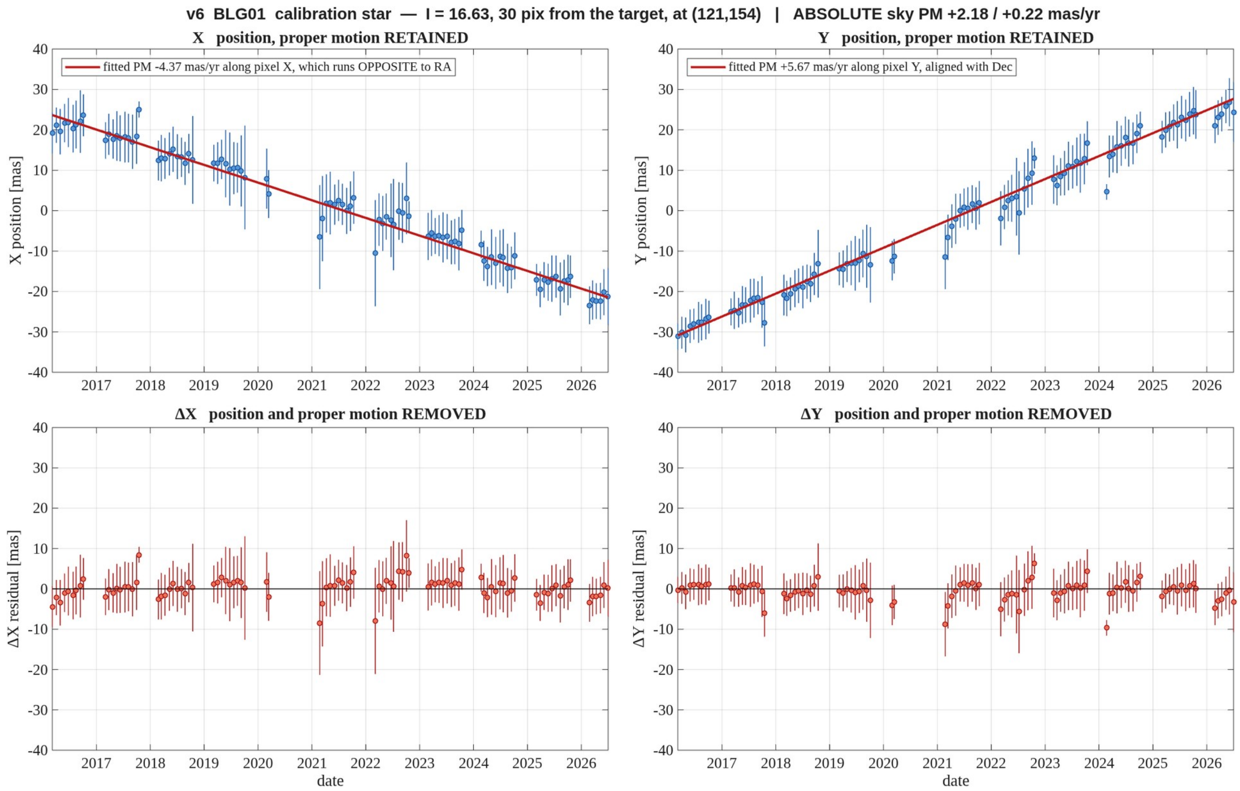

Figure 22. **BLG01** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d30.png`

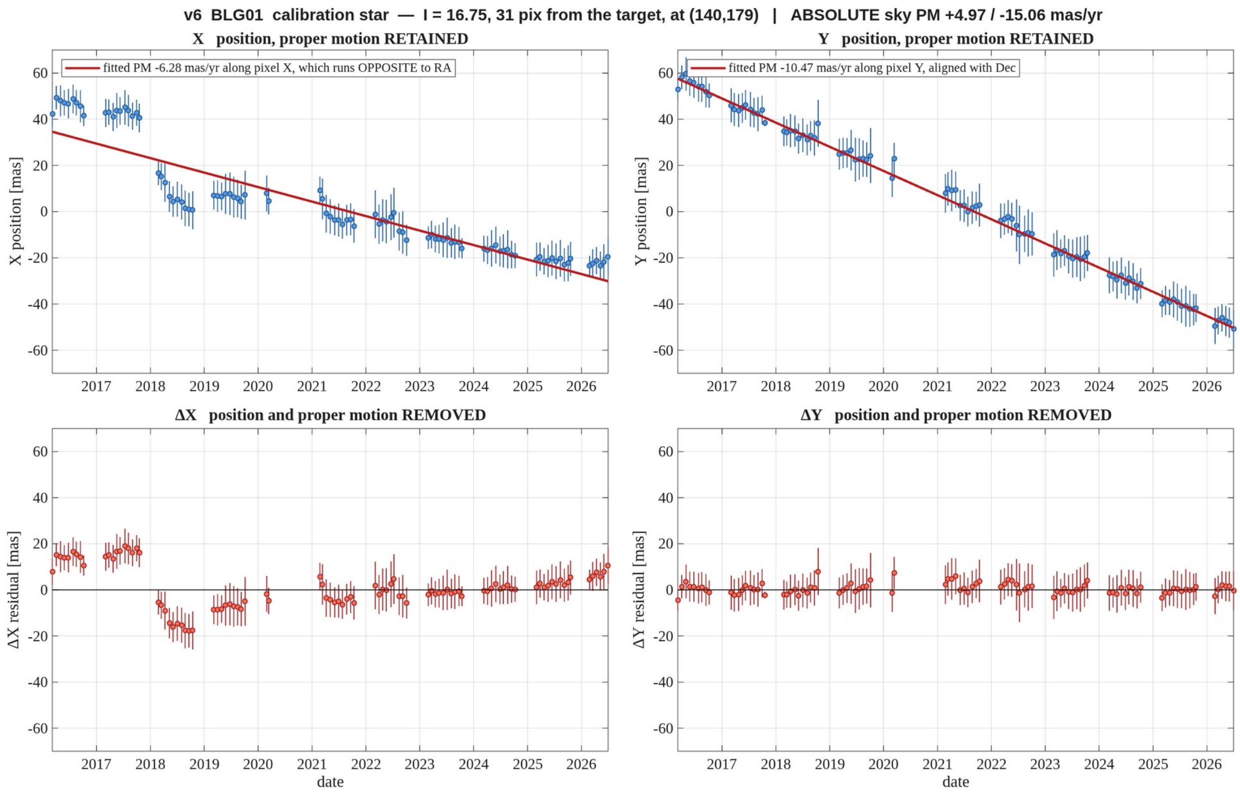

Figure 23. **BLG01** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d31.png`